```
from reportlab.lib.pagesizes
import A4
from reportlab.lib.styles
import ParagraphStyle
from reportlab.lib.units
import cm
from reportlab.lib.colors
import HexColor, white
from reportlab.platypus
import (SimpleDocTemplate,
Paragraph, Spacer,

     PageBreak, Table,
TableStyle, HRFlowable)
from reportlab.lib.enums
import TA_CENTER, TA_LEFT,
TA_JUSTIFY

NAVY  = HexColor('#0A1F44')
BLUE  = HexColor('#1E56D6')
LIGHT = HexColor('#F0F4FF')
```

```
GRAY  = HexColor('#666666')
LG    = HexColor('#CCCCCC')
GREEN = HexColor('#1F9D55')
GOLD  = HexColor('#E38A2C')
CODE_BG = HexColor('#1E1E2E')
CODE_FG = HexColor('#CDD6F4')
CODE_CM = HexColor('#6C7086')
WHITE = white

W, H = A4

def ps(name,
font='Helvetica', size=11,
color=HexColor('#333333'),
       bold=False,
align=TA_LEFT, leading=18,
before=0, after=8,
       left=0, bg=None,
italic=False):
    fn = 'Helvetica-Bold' if
bold else ('Helvetica-
Oblique' if italic else
'Helvetica')
```

```
    kwargs =
dict(fontName=fn,
fontSize=size,
textColor=color,

leading=leading,
spaceBefore=before,
spaceAfter=after,

leftIndent=left,
alignment=align)
    if bg:
kwargs['backColor'] = bg
    return
ParagraphStyle(name,
**kwargs)

body   = ps('body', size=11,
leading=18, after=8)
code   = ps('code', size=9,
color=CODE_FG, bg=CODE_BG,
leading=13, left=8, after=1,
before=1)
```

```
cmnt   = ps('cmnt', size=9,
color=CODE_CM, bg=CODE_BG,
leading=13, left=8, after=1)
footer = ps('footer', size=9,
color=GRAY, align=TA_CENTER)
note   = ps('note', size=10,
italic=True,
color=HexColor('#444'),
bg=LIGHT, left=10, before=4,
after=8)

def heading(text,
color=NAVY):
    d=[[Paragraph(text,
ps('h',bold=True,size=17,colo
r=WHITE,align=TA_CENTER))]]
    t=Table(d,colWidths=[W-
4*cm])

t.setStyle(TableStyle([('BACK
GROUND',(0,0),(-1,-1),color),
('PADDING',(0,0),
(-1,-1),12)]))
```

```
    return t

def topic_box(num, title):
    d=[[Paragraph(f'TOPIC
{num}',ps('tn',bold=True,size
=11,color=WHITE,align=TA_CENT
ER)),

Paragraph(title,ps('tt',bold=
True,size=13,color=NAVY))]]
    t=Table(d,colWidths=
[2.5*cm,W-6.5*cm])

t.setStyle(TableStyle([('BACK
GROUND',(0,0),(0,0),BLUE),
('BACKGROUND',(1,0),
(1,0),LIGHT),

('PADDING',(0,0),(-1,-1),10),
('VALIGN',(0,0),
(-1,-1),'MIDDLE')]))
    return t
```

```
def project_box(num, title):
    d=[[Paragraph(f'PROJECT
{num}',ps('pn',bold=True,size
=11,color=WHITE,align=TA_CENT
ER)),

Paragraph(title,ps('pt',bold=
True,size=13,color=WHITE))]]
    t=Table(d,colWidths=
[2.5*cm,W-6.5*cm])

t.setStyle(TableStyle([('BACK
GROUND',(0,0),(0,0),GREEN),
('BACKGROUND',(1,0),
(1,0),NAVY),

('TEXTCOLOR',(1,0),
(1,0),WHITE),

('PADDING',(0,0),(-1,-1),10),
('VALIGN',(0,0),
(-1,-1),'MIDDLE')]))
    return t
```

```
def code_block(lines):
    items =
[Paragraph('SOURCE CODE',
ps('cl',bold=True,size=9,colo
r=WHITE,bg=BLUE,align=TA_CENT
ER,before=8,after=0))]
    for line in lines:
        stripped =
line.strip()
        safe =
line.replace('&','&amp;').rep
lace('<','&lt;').replace('>',
'&gt;')
        if
stripped.startswith('#'):

items.append(Paragraph(safe,
cmnt))
        else:

items.append(Paragraph(safe,
code))
```

```
    items.append(Spacer(1,8))
    return items

def divider():
    return
HRFlowable(width='100%',thick
ness=1,color=LG,spaceAfter=6,
spaceBefore=6)

story = []

# ── COVER
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story.append(Spacer(1,2*cm))
d=
[[Paragraph('PYTHON',ps('ct',
bold=True,size=32,color=WHITE
,align=TA_CENTER))]]
c=Table(d,colWidths=[W-4*cm])
c.setStyle(TableStyle([('BACK
GROUND',(0,0),(-1,-1),NAVY),
('PADDING',(0,0),
```

```
(-1,-1),40)]))
story.append(c)
story.append(Spacer(1,0.4*cm)
)
d2=[[Paragraph('COMPLETE
SOURCE
CODE',ps('cs',size=15,color=W
HITE,align=TA_CENTER))]]
c2=Table(d2,colWidths=[W-
4*cm])
c2.setStyle(TableStyle([('BAC
KGROUND',(0,0),(-1,-1),BLUE),
('PADDING',(0,0),
(-1,-1),14)]))
story.append(c2)
story.append(Spacer(1,0.8*cm)
)
info=[['Platform','Code
Academy'],['Brand','SEA
TECHNOLOGIES MEDIA'],
      ['Instructor','Emmanuel
Adedayo'],
      ['Note','All code in
```

```
this PDF is exactly as taught
in the video'],
      ['Projects','Grade
Calculator + Python
Calculator'],

['Website','sealearninghub.on
render.com']]
it=Table([[Paragraph(k,ps('ik
',bold=True,size=11,color=NAV
Y)),Paragraph(v,body)] for
k,v in info],
         colWidths=[4.5*cm,W-
8.5*cm])
it.setStyle(TableStyle([('ROW
BACKGROUNDS',(0,0),(-1,-1),
[WHITE,LIGHT]),

('PADDING',(0,0),(-1,-1),10),
('GRID',(0,0),
(-1,-1),0.5,LG)]))
story.append(it)
story.append(Spacer(1,0.8*cm)
```

```
)
d3=[[Paragraph('Learn. Build.
Innovate.',ps('tag',bold=True
,size=16,color=WHITE,align=TA
_CENTER))]]
c3=Table(d3,colWidths=[W-
4*cm])
c3.setStyle(TableStyle([('BAC
KGROUND',(0,0),(-1,-1),GOLD),
('PADDING',(0,0),
(-1,-1),12)]))
story.append(c3)
story.append(PageBreak())

#
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# TOPIC 1 — Introduction
(print statements)
#
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```

```
==========
story.append(topic_box('1','Introduction to Python'))
story.append(Spacer(1,0.3*cm))
story.append(Paragraph('This is the first code from the video – using print() to display messages.',note))
story.extend(code_block([
    '# INTRODUCTION TO PYTHON',
    '# Code Academy | SEA TECHNOLOGIES MEDIA',
    '',
    'print("Hello World")',
    'print("Welcome to Code Academy")',
    'print("My name is Emmanuel Adedayo")',
    'print("Today we are learning Python")',
    'print("Starting with the
```

```
first thing \'PRINT\'")',
]))
story.append(PageBreak())

#
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# TOPIC 2 — Variables and
Data Types
#
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story.append(topic_box('2','V
ariables and Data Types'))
story.append(Spacer(1,0.3*cm)
)
story.extend(code_block([
    '# VARIABLES AND DATA
TYPES',
    '# Code Academy | SEA
TECHNOLOGIES MEDIA',
```

```
    '',
    '# STRING',
    'name = "Emmanuel"',
    'course = "python"',
    'platform = "code
academy"',
    '',
    '# INTEGER',
    'age = 20',
    'students = 500',
    '',
    '# FLOAT',
    'price = 9.99',
    'height = 5.9',
    'percentage = 98.5',
    '',
    '# BOOLEAN',
    'is_free = True',
    'is_paid = False',
    'is_active = True',
    '',
    '# Print all of them',
    'print(name)',
```

```
    'print(course)',
    'print(platform)',
    'print(age)',
    'print(students)',
    'print(price)',
    'print(height)',
    'print(percentage)',
    'print(is_free)',
    'print(is_paid)',
    'print(is_active)',
    '',
    '# Check the datatype of
any variable',
    'print(type(name))',
    'print(type(course))',
    'print(type(platform))',
    'print(type(age))',
    'print(type(students))',
    'print(type(price))',
    'print(type(height))',

'print(type(percentage))',
    'print(type(is_free))',
```

```
    'print(type(is_paid))',
    'print(type(is_active))',
]))
story.append(PageBreak())

# ═══════════════════════════════════════════════════════════
# TOPIC 3 — Operators and Input
# ═══════════════════════════════════════════════════════════
story.append(topic_box('3','Operators and Input'))
story.append(Spacer(1,0.3*cm))
story.extend(code_block([
    '# ARITHMETIC OPERATORS',
    '# Code Academy | SEA TECHNOLOGIES MEDIA',
```

```
    '',
    'a = 20',
    'b = 6',
    '',
    'print(a + b)   #
ADDITION       — 26',
    'print(a - b)   #
SUBTRACTION    — 14',
    'print(a * b)   #
MULTIPLICATION — 120',
    'print(a / b)   #
DIVISION       — 3.333',
    'print(a // b)  # FLOOR
DIVISION — 3',
    'print(a ** b)  #
POWER          — 64000000',
    'print(a % b)   #
MODULUS        — 2',
    '',
    'print()',
    'print()',
    '',
    '# COMPARISON OPERATORS',
```

```
    'print(a > b)   # True',
    'print(a < b)   # False',
    'print(a == b)  # False',
    'print(a != b)  # True',
    'print(a >= b)  # True',
    'print(a <= b)  # False',
    '',
    'print()',
    'print()',
    '',
    '# INPUT',
    'name   = input("What is
your name   ")',
    'course = input("Which
course do you like the most
on this platform  ")',
    'age    = input("How old
are you when you started this
course  ")',
    '',
    'print("Hello " + name +
" You\'re welcome to this
python course and am glad you
```

```
love " + course +',
    '      " the most, the
founder will be glad to hear
that. You\'re one of the
youngest age " + age + "
years old on this
platform")',
    '',
    '# BUILDING A SIMPLE
CALCULATOR',
    'print()',
    'print("\\n---SIMPLE
CALCULATOR---")',
    'num1 =
float(input("Enter first
number  "))',
    'num2 =
float(input("Enter second
number  "))',
    '',
    'print("Addition:       "
+ str(int(num1 + num2)))',
    'print("Subtraction:    "
```

```
+ str(int(num1 - num2)))',
    'print("Multiplication: "
+ str(int(num1 * num2)))',
    'print("Division:       "
+ str(int(num1 / num2)))',
]))
story.append(PageBreak())

#
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# TOPIC 4 — Conditions
#
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story.append(topic_box('4','C
onditions (if, elif, else)'))
story.append(Spacer(1,0.3*cm)
)
story.extend(code_block([
    '# CONDITIONS — if, elif,
```

```
else',
    '# Code Academy | SEA
TECHNOLOGIES MEDIA',
    '',
    '# BASIC IF ELSE',
    'years = 18',
    '',
    'if years >= 18:',
    '
print("Congratulation You are
an Adult")',
    'else:',
    '    print("Sorry You are
not an Adult yet")',
    '',
    '',
    '# IF ELIF ELSE',
    'score = 75',
    '',
    'if score >= 70:',
    '
print("Distinction")',
    'elif score >= 50:',
```

```
    '    print("Pass")',
    'elif score >= 40:',
    '    print("Nearly
Pass")',
    'else:',
    '    print("Fail")',
    '',
    '',
    '# WITH USER INPUT',
    'print("\\n--- GRADE
CHECKER ---")',
    'score =
float(input("Enter your
score: "))',
    '',
    'if score >= 70:',
    '    print("Excellent!
You got a Distinction")',
    'elif score >= 50:',
    '    print("Good job! You
Passed")',
    'elif score >= 40:',
    '    print("Almost there!
```

```
Nearly Pass")',
    'else:',
    '    print("Do not give
up! Keep studying")',
    '',
    '',
    '# LOGIN SYSTEM',
    'print("\\n--- LOGIN
SYSTEM ---")',
    'password = input("Enter
password: ")',
    '',
    'if password ==
"python123":',
    '    print("Access
granted! Welcome")',
    'else:',
    '    print("Wrong
password. Access denied")',
]))
story.append(PageBreak())

#
```

```
# ═══════════════════════════════════════════════════════════
# TOPIC 5 — Loops
# ═══════════════════════════════════════════════════════════
story.append(topic_box('5','Loops (for loop and while loop)'))
story.append(Spacer(1,0.3*cm))
story.extend(code_block([
    '# LOOPS — for loop and while loop',
    '# Code Academy | SEA TECHNOLOGIES MEDIA',
    '',
    '# FOR LOOP — basic counting',
    'for i in range(5):',
    '    print(i)',
```

```
    '',
    '',
    '# FOR LOOP – with start
and end',
    'for i in range(1, 12):',
    '    print("Number: " +
str(i))',
    '',
    '',
    '# FOR LOOP – through a
tuple',
    'courses = ("HTML",
"CSS", "JAVASCRIPT",
"PYTHON")',
    'for course in courses:',
    '    print("I am learning
" + course)',
    '',
    '',
    '# WHILE LOOP',
    'count = 1',
    'while count <= 100:',
    '    print("Count is: " +
```

```
str(count))',
    '    count = count + 1',
    '',
    '',
    '# REAL WORLD — 
Multiplication Table',
    'print("\\n--- 
MULTIPLICATION TABLE ---")',
    'number = 
int(input("Enter a number: 
"))',
    'for i in range(1, 11):',
    '    print(str(number) + 
" x " + str(i) + " = " + 
str(number * i))',
]))
story.append(PageBreak())

# 
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# TOPIC 6 — Functions
```

```
#
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story.append(topic_box('6','F
unctions'))
story.append(Spacer(1,0.3*cm)
)
story.extend(code_block([
    '# FUNCTIONS',
    '# Code Academy | SEA
TECHNOLOGIES MEDIA',
    '',
    '# BASIC FUNCTION — no
parameters',
    'def greet():',
    '    print("Hello,
welcome to Code Academy")',
    '',
    '# Call the function
three times',
    'greet()',
    'greet()',
```

```
    'greet()',
    '',
    '',
    '# FUNCTION WITH
PARAMETER',
    'def
greet_students(name):',
    '    print("Hello " +
name + " Welcome to
python!")',
    '',

'greet_students("Emmanuel")',
    'greet_students("Dave")',
    'greet_students("John")',
    '',
    '',
    '# REAL WORLD FUNCTION –
with return value',
    'def
calculate_area(length,
width):',
    '    area = length *
```

```
width',
    '    return area',
    '',
    'room1 =
calculate_area(10, 5)',
    'room2 =
calculate_area(8, 6)',
    'print("Room 1 area: " +
str(room1) + " square
meters")',
    'print("Room 2 area: " +
str(room2) + " square
meters")',
]))
story.append(PageBreak())

#
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# TOPIC 7 — Lists and Tuples
#
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```

```
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story.append(topic_box('7','Lists and Tuples'))
story.append(Spacer(1,0.3*cm))
story.extend(code_block([
    '# LISTS AND TUPLES',
    '# Code Academy | SEA TECHNOLOGIES MEDIA',
    '',
    '# CREATING LISTS',
    'students = ["Emmanuel", "Dave", "John", "Jane", "Mike"]',
    'courses  = ["HTML", "CSS", "JavaScript", "Python", "Git", "Django"]',
    'scores   = [85, 92, 78, 65, 90]',
    '',
    '# Accessing items (index starts at 0)',
```

```
    'print(students[0])   #
Emmanuel — first item',
    'print(courses[1])    #
CSS — second item',
    'print(scores[-1])    #
90 — last item',
    '',
    '# Getting the length',
    'print(len(students))  #
5',
    '',
    '# UPDATING AN ITEM',
    'students[0] = "Emmanuel
Adedayo"',
    'print(students)',
    '',
    '',
    '# TUPLE — cannot be
changed after creation',
    'days = ("Monday",
"Tuesday", "Wednesday",',
    '        "Thursday",
"Friday", "Saturday",
```

```
"Sunday")',
    'print(days)',
]))
story.append(PageBreak())

#
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# TOPIC 8 — Dictionaries
#
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story.append(topic_box('8','D
ictionaries'))
story.append(Spacer(1,0.3*cm)
)
story.extend(code_block([
    '# DICTIONARIES',
    '# Code Academy | SEA
TECHNOLOGIES MEDIA',
    '',
```

```
    '# CREATING A
DICTIONARY',
    'students = {',
    '    "name": "Emmanuel
Adedayo",',
    '    "age": 20,',
    '    "course":
"python",',
    '    "email":
"seatechnologiesmedia@gmail.c
om",',
    '    "is_active": True',
    '}',
    '',
    '# Accessing values',

'print(students["name"])',
    'print(students["age"])',

'print(students["course"])',

'print(students["email"])',
```

```
'print(students["is_active"])
',
    '',
    '# CHECKING IF A KEY
EXISTS',
    'if "email" in
students:',
    '    print("email found "
+ students["email"])',
    '',
    '',
    '# DICTIONARY OF
DICTIONARIES',
    'students = {',
    '    "student1": {',
    '        "name":
"Emmanuel",',
    '        "score": 85,',
    '        "grade":
"Distinction"',
    '    },',
    '    "student2": {',
    '        "name":
```

```
"Dave",',
    '        "score": 72,',
    '        "grade":
"Distinction"',
    '    },',
    '    "student3": {',
    '        "name":
"John",',
    '        "score": 55,',
    '        "grade":
"Pass"',
    '    }',
    '}',
    '',
    'print("\\n--- ALL
STUDENTS ---")',
    'for student_id, info in
students.items():',
    '    print(info["name"] +
" — " + info["grade"])',
]))
story.append(PageBreak())
```

```
#
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# PROJECT 1 — Grade
Calculator
#
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story.append(project_box('1',
'Grade Calculator'))
story.append(Spacer(1,0.3*cm)
)
story.append(Paragraph(
    'Complete project code —
uses functions, loops,
conditions, lists and file
handling.',note))
story.extend(code_block([
    '#
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```

```
    '# GRADE CALCULATOR',
    '# Code Academy | SEA
TECHNOLOGIES MEDIA',
    '# Built with Python',
    '#
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    '',
    'def
get_grade(average):',
    '    if average >= 70:',
    '        return
"DISTINCTION"',
    '    elif average >=
50:',
    '        return "PASS"',
    '    elif average >=
40:',
    '        return "NEARLY
PASS"',
    '    else:',
    '        return "FAIL"',
    '',
```

```
    'def
get_remark(average):',
    '    if average >= 70:',
    '        return
"Excellent performance! Keep
it up!"',
    '    elif average >=
50:',
    '        return "Good
job! You can do even
better!"',
    '    elif average >=
40:',
    '        return "Almost
there! Study harder next
time!"',
    '    else:',
    '        return "Do not
give up! Keep pushing!"',
    '',
    'def save_result(name,
subjects, scores, average,
grade):',
```

```
    '    with
open("results.txt", "a") as
file:',
    '        file.write("=" *
40 + "\\n")',
    '
file.write("Name: " + name +
"\\n")',
    '        for i in
range(len(subjects)):',
    '
file.write(subjects[i] + ": "
+ str(scores[i]) + "\\n")',
    '
file.write("Average: " +
str(round(average, 2)) +
"\\n")',
    '
file.write("Grade: " + grade
+ "\\n")',
    '        file.write("=" *
40 + "\\n")',
    '',
```

```
    'def main():',
    '    print("=" * 45)',
    '    print("      CODE
ACADEMY GRADE CALCULATOR")',
    '    print("      SEA
TECHNOLOGIES MEDIA")',
    '    print("=" * 45)',
    '',
    '    name =
input("\\nEnter student full
name: ")',
    '',
    '    subjects = [',
    '        "Mathematics",',
    '        "English
Language",',
    '        "Python
Programming",',
    '        "Web
Development",',
    '        "Problem
Solving"',
    '    ]',
```

```
    '',
    '    scores = []',
    '    print("\\nEnter
scores for each subject (0-
100):\\n")',
    '',
    '    for subject in
subjects:',
    '        while True:',
    '            try:',
    '                score =
float(input(subject + ":
"))',
    '                if 0 <=
score <= 100:',
    '
scores.append(score)',
    '
break',
    '                else:',
    '
print("Score must be between
0 and 100")',
```

```
    '            except
ValueError:',
    '
print("Please enter a valid
number")',
    '',
    '    total   =
sum(scores)',
    '    average = total /
len(scores)',
    '    grade   =
get_grade(average)',
    '    remark  =
get_remark(average)',
    '',
    '    print("\\n" + "=" *
45)',
    '    print("
RESULT FOR " +
name.upper())',
    '    print("=" * 45)',
    '',
    '    for i in
```

```
range(len(subjects)):',
    '        print(f"
{subjects[i]:<25}
{scores[i]}")',
    '',
    '    print("-" * 45)',
    '    print(f"{\'Total:\':
<25} {total}")',
    '    print(f"
{\'Average:\':<25}
{round(average, 2)}")',
    '    print(f"{\'Grade:\':
<25} {grade}")',
    '    print("-" * 45)',
    '    print(f"Remark:
{remark}")',
    '    print("=" * 45)',
    '',
    '    save_result(name,
subjects, scores, average,
grade)',
    '    print("\\nResult
saved to results.txt
```

```
successfully!")',
    '    print("\\nThank you
for using Code Academy Grade
Calculator!")',
    '    print("Learn. Build.
Innovate.")',
    '',
    '# Run the program',
    'main()',
]))
story.append(PageBreak())

#
═══════════════════════════════
═══════════════════════════════
═══════
# PROJECT 2 — Python
Calculator
#
═══════════════════════════════
═══════════════════════════════
═══════
story.append(project_box('2',
```

```
'Python Calculator'))
story.append(Spacer(1,0.3*cm)
)
story.append(Paragraph(
    'Complete calculator
project — uses functions,
while loop, conditions, lists
and input validation.',note))
story.extend(code_block([
    '#
═══════════════════════════
═══════════════════',
    '# PYTHON CALCULATOR',
    '# Code Academy | SEA
TECHNOLOGIES MEDIA',
    '# Built with Python
only',
    '#
═══════════════════════════
═══════════════════',
    '',
    '# ── FUNCTIONS
──────────────────────────
```

```
—',
    '',
    'def add(a, b):',
    '    return a + b',
    '',
    'def subtract(a, b):',
    '    return a - b',
    '',
    'def multiply(a, b):',
    '    return a * b',
    '',
    'def divide(a, b):',
    '    if b == 0:',
    '        return "Error:
Cannot divide by zero"',
    '    return a / b',
    '',
    'def power(a, b):',
    '    return a ** b',
    '',
    'def modulus(a, b):',
    '    if b == 0:',
    '        return "Error:
```

```
Cannot divide by zero"',
    '    return a % b',
    '',
    'def square_root(a):',
    '    if a < 0:',
    '        return "Error:
Cannot find square root of
negative number"',
    '    return a ** 0.5',
    '',
    '',
    '# ── DISPLAY MENU
──────────────────────────',
    '',
    'def show_menu():',
    '    print("\\n" + "=" *
40)',
    '    print("        SEA
TECH CALCULATOR")',
    '    print("        Code
Academy | Python")',
    '    print("=" * 40)',
    '    print("  1.
```

```
Addition          ( + )")',
    '    print("  2.
Subtraction       ( - )")',
    '    print("  3.
Multiplication    ( * )")',
    '    print("  4.
Division          ( / )")',
    '    print("  5.
Power             ( ^ )")',
    '    print("  6.
Modulus           ( % )")',
    '    print("  7.  Square
Root       ( v )")',
    '    print("  8.
Calculation History")',
    '    print("  9.  Clear
History")',
    '    print("  0.
Exit")',
    '    print("=" * 40)',
    '',
    '',
    '# ── GET NUMBER FROM
```

```
USER ————————————————————',
    '',
    'def
get_number(prompt):',
    '    while True:',
    '        try:',
    '            number =
float(input(prompt))',
    '            return
number',
    '        except
ValueError:',
    '            print("
Invalid input. Please enter a
number.")',
    '',
    '',
    '# —— GET CHOICE FROM
USER ————————————————————',
    '',
    'def get_choice():',
    '    while True:',
    '        try:',
```

```
    '            choice =
int(input("\\n  Enter your
choice: "))',
    '            if 0 <=
choice <= 9:',
    '                return
choice',
    '            else:',
    '                print("
Please enter a number between
0 and 9.")',
    '        except
ValueError:',
    '            print("
Invalid input. Please enter a
number.")',
    '',
    '',
    '# ── FORMAT RESULT
──────────────────────────',
    '',
    'def
format_number(num):',
```

```
    '    if isinstance(num,
str):',
    '        return num',
    '    if num ==
int(num):',
    '        return
str(int(num))',
    '    return
str(round(num, 6))',
    '',
    '',
    '# ── SHOW HISTORY
──────────────────────────────',
    '',
    'def
show_history(history):',
    '    print("\\n" + "=" *
40)',
    '    print("
CALCULATION HISTORY")',
    '    print("=" * 40)',
    '    if len(history) ==
0:',
```

```
    '        print("  No
calculations yet.")',
    '    else:',
    '        for i, record in
enumerate(history):',
    '            print(f"  {i
+ 1}. {record}")',
    '    print("=" * 40)',
    '',
    '',
    '# ── MAIN PROGRAM
──────────────────────────',
    '',
    'def main():',
    '    history = []',
    '',
    '    print("\\n" + "=" *
40)',
    '    print("   Welcome to
SEA TECH Calculator")',
    '    print("   Code
Academy | SEA TECHNOLOGIES
MEDIA")',
```

```
    '    print("=" * 40)',
    '',
    '    while True:',
    '        show_menu()',
    '        choice =
get_choice()',
    '',
    '        if choice ==
0:',
    '            print("\\n"
+ "=" * 40)',
    '            print("
Thank you for using SEA TECH
Calculator!")',
    '            print("
Learn. Build. Innovate.")',
    '            print("=" *
40 + "\\n")',
    '            break',
    '',
    '        elif choice ==
8:',
    '
```

```
show_history(history)',
    '            continue',
    '',
    '        elif choice ==
9:',
    '
history.clear()',
    '            print("\\n
History cleared
successfully.")',
    '            continue',
    '',
    '        elif choice ==
7:',
    '            a =
get_number("\\n  Enter
number: ")',
    '            result =
square_root(a)',
    '            record =
f"sqrt({format_number(a)}) =
{format_number(result)}"',
    '            if not
```

```
isinstance(result, str):',
    '
history.append(record)',
    '            print(f"\\n
Result:
{format_number(result)}")',
    '',
    '        else:',
    '            a =
get_number("\\n  Enter first
number:  ")',
    '            b =
get_number("  Enter second
number: ")',
    '',
    '            if choice ==
1:',
    '                result =
add(a, b)',
    '                symbol =
"+"',
    '            elif choice
== 2:',
```

```
    '                result =
subtract(a, b)',
    '                symbol =
"-"',
    '            elif choice
== 3:',
    '                result =
multiply(a, b)',
    '                symbol =
"x"',
    '            elif choice
== 4:',
    '                result =
divide(a, b)',
    '                symbol =
"/"',
    '            elif choice
== 5:',
    '                result =
power(a, b)',
    '                symbol =
"^"',
    '            elif choice
```

```
== 6:',
    '                result =
modulus(a, b)',
    '                symbol =
"%"',
    '',
    '            record = (f"
{format_number(a)} {symbol}
"',
    '                      f"
{format_number(b)} =
{format_number(result)}")',
    '',
    '            if not
isinstance(result, str):',
    '
history.append(record)',
    '',
    '            print(f"\\n
Result:
{format_number(result)}")',
    '',
    '        print("\\n" + "-
```

```
" * 40)',
    '        again = input("
Calculate again? (yes / no):
").lower().strip()',
    '        if again !=
"yes" and again != "y":',
    '            print("\\n"
+ "=" * 40)',
    '            print("
Thank you for using SEA TECH
Calculator!")',
    '            print("
Learn. Build. Innovate.")',
    '            print("=" *
40 + "\\n")',
    '            break',
    '',
    '',
    '# ── RUN
──────────────────────────────
─────────',
    'main()',
]))
```

```
story.append(PageBreak())

# ── FINAL PAGE
──────────────────────────────────
──────────────────────
story.append(Spacer(1,2*cm))
fd=
[[Paragraph('Congratulations!
',ps('cg',bold=True,size=24,c
olor=WHITE,align=TA_CENTER))]
]
ft=Table(fd,colWidths=[W-
4*cm])
ft.setStyle(TableStyle([('BAC
KGROUND',(0,0),
(-1,-1),GREEN),('PADDING',
(0,0),(-1,-1),25)]))
story.append(ft)
story.append(Spacer(1,0.5*cm)
)
story.append(Paragraph(
    'You have completed the
Python course on Code
```

```
Academy. '
    'Study this source code,
practice it, modify it and
build on top of it.',

ps('fm',size=13,align=TA_CENT
ER,leading=20)))
story.append(Spacer(1,0.8*cm)
)
tgd=[[Paragraph('Learn.
Build.
Innovate.',ps('tg',bold=True,
size=18,color=WHITE,align=TA_
CENTER))]]
tgt=Table(tgd,colWidths=[W-
4*cm])
tgt.setStyle(TableStyle([('BA
CKGROUND',(0,0),
(-1,-1),GOLD),('PADDING',
(0,0),(-1,-1),14)]))
story.append(tgt)
story.append(Spacer(1,0.3*cm)
)
```

```
story.append(Paragraph(
    'Emmanuel Adedayo |
Founder & Lead Instructor |
Code Academy | SEA
TECHNOLOGIES MEDIA',footer))
story.append(Paragraph('seale
arninghub.onrender.com',foote
r))

doc=SimpleDocTemplate('/mnt/u
ser-data/outputs/python-
course/python-source-
code.pdf',

pagesize=A4,rightMargin=2*cm,
leftMargin=2*cm,

topMargin=2*cm,bottomMargin=2
*cm)
doc.build(story)
print("Source Code PDF
done!")
```